

PAPER Component Activation, Waste-Class Routing, and Decommissioning of a Low-Activation Compact Spherical-Tokamak Breeder: No Isotope Beyond a Century's Half-Life

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Abstract

The environmental case for a fusion breeder is settled at end of life, in the activation it accumulates, the waste classes that inventory passes through as it cools, and the ease with which the machine is retired. We report a rigorous activation, waste-routing, and decommissioning account for the Kronos *Hyperion* breeder — a compact negative-triangularity spherical tokamak operating at $P_{\text{fus}} = 85.04$ MW, $I_p = 9.66$ MA, $B_0 = 8$ T, with a 16.84 T high-temperature-superconducting centrepost. A first-wall neutron flux of 5.628×10^{14} n cm⁻² s⁻¹ at a wall load of 1.89 MW m⁻² is tallied on a 175-group VITAMIN-J structure in a three-dimensional OpenMC model normalized to a fusion neutron source of $S_n = 3.02 \times 10^{19}$ ns⁻¹, and the resulting spectra drive a full-chain Bateman transmutation–decay inventory in ALARA/FISPACT-II with per-component classification against the U.S. 10 CFR 61 waste rule. The result is unambiguous: at a two-full-power-year exposure *every activated component classifies at or below Class C, with no greater-than-Class-C (GTCC) waste*; the largest Class-C index anywhere in the build is the ceramic breeder at 9.4×10^{-3} , two orders of magnitude below the regulatory boundary, and only tritium retention pushes any stream above Class A. The qualified low-activation high-entropy-alloy structure follows a clean descent — intermediate-level at one year, low-level at ten, and clearable below regulatory concern at one hundred — with a *long-lived isotopic fraction of zero at every cooling time*, because the composition screen categorically excludes the ⁹⁴Nb, ⁹³Zr and molybdenum activation channels that keep conventional alloys intermediate-level indefinitely. The retired inventory is passively safe: decay heat is ≤ 62 W kg⁻¹ at one year, two orders below fission post-shutdown densities with no melt-driving pathway, and the stored energies are small — 16.06 MJ of plasma thermal energy (≈ 3.8 kg TNT-equivalent) and a radiologically inert ~ 0.3 GJ of coil field energy. Decommissioning handles a modest, largely recyclable footprint of order 40 t of steel and 90 m³ of concrete per installed megawatt at a lifecycle intensity of 10–30 gCO₂e kWh⁻¹. The clean end of life is not a mitigation bolted on afterward; it is a computed consequence of a deliberately clean material choice, and it closes.

Keywords: activation inventory; radioactive-waste routing; 10 CFR 61 classification; clearance; decommissioning; low-activation high-entropy alloys; rigorous two-step; lifecycle carbon; spherical-tokamak breeder

1 Introduction

The public and regulatory case for fusion is decided less by the peak performance a machine reaches than by what it leaves behind. A device that produced abundant neutrons but a legacy of long-lived, deep-disposal waste would forfeit much of the advantage that motivates fusion in the first place. The activation inventory of the in-vessel components, the routing of that inventory through waste classes as it cools, and the ease with which the retired machine can be dismantled and its material recycled are therefore first-order design outputs, not post-hoc environmental book-keeping [2, 4]. This is doubly true for a *compact* breeder intended for near-urban, distributed deployment, where the licensing posture and the decommissioning burden are gating commercial constraints [39].

The Kronos *Hyperion* breeder is a compact, high-field, negative-triangularity spherical tokamak whose canonical operating point is fixed by the Tier-2 design freeze at major radius $R_0 = 1.20$ m, aspect ratio $A = 2.5$, elongation $\kappa = 2.0$, triangularity $\delta = -0.30$, on-axis field $B_0 = 8$ T, plasma current $I_p = 9.66$ MA and fusion gain $Q = 3.076$, producing $P_{\text{fus}} = 85.04$ MW with a 16.84 T peak field on a demountable, no-insulation REBCO centrepost [1, 28, 29]. The machine is designed as a

strategic-materials platform — a 14 MeV neutron source for tritium breeding, helium-3 co-production and medical-isotope manufacture — rather than as a net-electric plant [32]. In that role the fast-neutron field that loads and activates every in-vessel component is the same field that breeds the fuel, so the activation and waste account is inseparable from the mission.

Prior art. Activation and waste classification for magnetic fusion has a mature methodology. The transmutation–decay inventory is solved from tabulated cross-sections and decay data with inventory codes such as FISPACT-II [6] and ALARA [7], driven by neutron spectra from Monte Carlo transport (OPENMC [8], coupled to CAD geometry through DAGMC [10]) on evaluated libraries such as ENDF/B-VIII.0 [9]. The European DEMO programme has established that reduced-activation ferritic–martensitic (RAFM) steels such as EUROFER-97 reach low-level or clearable inventories on century timescales provided impurity control suppresses the long-lived activation products [2, 3, 16, 17], and that the residual waste class is frequently set not by the bulk structure but by trace elements and by tritium management [2, 5]. The waste-classification arithmetic itself is codified in the U.S. Nuclear Regulatory Commission’s 10 CFR 61.55 sum-of-fractions rule [13, 14] and, for the clearable endpoint, in the IAEA clearance levels [15]. What has been missing for a compact, high-wall-load spherical-tokamak breeder is a transport-driven, per-component account tied to a frozen engineering design point and to a specific, first-principles-screened low-activation alloy family.

Contribution. This paper supplies that account. Our contributions are: (i) a self-contained derivation of the governing activation transport, transmutation–decay and waste-classification equations, cast as the formal problem solved here (Sec. 2); (ii) a three-dimensional, spectrum-resolved rigorous-two-step (R2S) computational methodology run on the Kronos NVIDIA GPU high-performance-computing (HPC) campaign, with explicit normalization, discretization and reproducibility controls (Sec. 3); (iii) a per-component activation and waste-class result at a two-full-power-year exposure showing no GTCC waste and a largest Class-C index of 9.4×10^{-3} (Sec. 4); (iv) a waste-routing timeline for the qualified high-entropy-alloy (HEA) structure — ILW at one year, LLW at ten, clearable at one hundred, long-lived fraction zero — contrasted quantitatively against dirty reference alloys; and (v) the passive-safety, stored-energy, fuel-footprint and lifecycle-carbon envelope that closes the decommissioning case. The alloy design that furnishes the clearable endpoint is developed in companion first-principles studies [33, 34]; the spatially-resolved shutdown-dose twin and streaming/occupational-dose neighbours are reported separately [35, 36]; the displacement-damage lifetime that sets the component-replacement cadence, and hence the mass flow through the decommissioning stream, is quantified in the first-wall and centrepost-life studies [37]; the tritium source term and accident case that dominate the operational (as opposed to end-of-life) radiological hazard are treated in the environmental and tritium-safety companion [38].

2 Governing equations and the formal problem

The activation and waste problem couples three physics layers: neutron transport, which fixes the local flux and spectrum; transmutation–decay kinetics, which propagate the nuclide inventory through irradiation and cooling; and the regulatory classification functionals, which map that inventory to a waste class and a clearance time. We state each in turn and then define the formal problem.

2.1 Neutron transport and the group flux

The neutron field obeys the steady-state linear Boltzmann transport equation for the angular flux $\psi(\mathbf{r}, \boldsymbol{\Omega}, E)$,

$$\boldsymbol{\Omega} \cdot \nabla \psi + \Sigma_t(\mathbf{r}, E) \psi = \int_{4\pi} \int_0^\infty \Sigma_s(\mathbf{r}, \boldsymbol{\Omega}' \rightarrow \boldsymbol{\Omega}, E' \rightarrow E) \psi(\mathbf{r}, \boldsymbol{\Omega}', E') dE' d\boldsymbol{\Omega}' + S(\mathbf{r}, \boldsymbol{\Omega}, E), \quad (1)$$

with total and scattering macroscopic cross-sections Σ_t, Σ_s and a fixed fusion source $S = \frac{1}{4\pi} s(\mathbf{r}) \chi(E)$ whose energy spectrum $\chi(E)$ is the 14.06 MeV D–T birth line broadened by the ion temperature. The quantity the activation calculation requires is the multigroup scalar flux, obtained by integrating ψ over angle and over the g -th energy bin of a group structure $\{E_g\}_{g=0}^G$,

$$\phi_g(\mathbf{r}) = \int_{E_g}^{E_{g+1}} \int_{4\pi} \psi(\mathbf{r}, \boldsymbol{\Omega}, E) d\boldsymbol{\Omega} dE, \quad g = 1, \dots, G. \quad (2)$$

We solve Eq. (1) by Monte Carlo, which estimates ϕ_g as a track-length tally over simulated histories; the estimator is unbiased and its statistical error scales as $N_{\text{hist}}^{-1/2}$. The tally is returned per source neutron and is placed on an absolute basis by the fusion source rate. Charged fusion products deposit their energy in the plasma, so the neutron source rate follows directly from the fusion power and the D–T energy release $Q_{\text{DT}} = 17.59$ MeV:

$$S_n = \frac{P_{\text{fus}}}{Q_{\text{DT}}} = \frac{85.04 \text{ MW}}{17.59 \text{ MeV}} = 3.02 \times 10^{19} \text{ ns}^{-1}, \quad (3)$$

one 14 MeV neutron per reaction. The absolute group flux is then $\Phi_g(\mathbf{r}) = S_n \phi_g(\mathbf{r})$, and the first-wall total flux evaluated on this basis is $\Phi_{\text{FW}} = 5.628 \times 10^{14} \text{ n cm}^{-2} \text{ s}^{-1}$, of which $1.071 \times 10^{14} \text{ n cm}^{-2} \text{ s}^{-1}$ lies above 10 MeV and 17.1% sits in the 14 MeV band.

2.2 Transmutation–decay kinetics: the Bateman system

Under the spectrum-averaged flux the nuclide number densities $N_i(\mathbf{r}, t)$ evolve by the transmutation–decay (Bateman) system, in which each species is fed by decay and neutron-induced production from its precursors and depleted by its own decay and neutron absorption,

$$\frac{dN_i}{dt} = \sum_{j \neq i} \left(\lambda_{j \rightarrow i} + \bar{\phi} \bar{\sigma}_{j \rightarrow i} \right) N_j - \left(\lambda_i + \bar{\phi} \bar{\sigma}_i \right) N_i, \quad \bar{\phi} \bar{\sigma}_{j \rightarrow i} = \sum_g \phi_g \sigma_{j \rightarrow i}^g, \quad (4)$$

where λ_i is the decay constant, $\lambda_{j \rightarrow i}$ the branching-weighted decay rate from j to i , $\sigma_{j \rightarrow i}^g$ the group reaction cross-section (capture, $(n, 2n)$, (n, p) , (n, α) , $\dots$), and the one-group collapse uses the transport spectrum of Eq. (2). Collecting the inventory into a vector $\mathbf{N} = (N_1, \dots, N_M)^\top$, Eq. (4) is the linear autonomous system

$$\frac{d\mathbf{N}}{dt} = \mathbf{A} \mathbf{N}, \quad \mathbf{A} = \mathbf{A}_{\text{dec}} + \bar{\phi} \mathbf{A}_{\text{tr}}, \quad (5)$$

with the sparse transmutation–decay matrix $\mathbf{A}$ carrying decay on $\mathbf{A}_{\text{dec}}$ and flux-driven transmutation on $\mathbf{A}_{\text{tr}}$. Its formal solution over a constant-flux interval is the matrix exponential

$$\mathbf{N}(t) = e^{\mathbf{A}t} \mathbf{N}(0). \quad (6)$$

A schedule of irradiation and cooling is composed by chaining such intervals: during irradiation $\bar{\phi} > 0$ so $\mathbf{A} = \mathbf{A}_{\text{dec}} + \bar{\phi} \mathbf{A}_{\text{tr}}$; during cooling $\bar{\phi} = 0$ so $\mathbf{A} = \mathbf{A}_{\text{dec}}$ and the inventory relaxes purely by decay,

$$\mathbf{N}(t_{\text{irr}} + \tau) = e^{\mathbf{A}_{\text{dec}} \tau} e^{(\mathbf{A}_{\text{dec}} + \bar{\phi} \mathbf{A}_{\text{tr}}) t_{\text{irr}}} \mathbf{N}(0). \quad (7)$$

2.3 Activity, decay heat, and the passive-safety condition

The specific activity and the specific decay heat of a component of mass m follow from the inventory as

$$a(t) = \frac{1}{m} \sum_i \lambda_i N_i(t), \quad \dot{q}(t) = \frac{1}{m} \sum_i \lambda_i N_i(t) \bar{E}_i, \quad (8)$$

where $\bar{E}_i = \bar{E}_i^\alpha + \bar{E}_i^\beta + \bar{E}_i^\gamma$ is the mean energy released per disintegration. The passive-safety (“walk-away”) requirement is that the retired inventory cannot heat itself to a damage temperature faster than it can shed heat. Writing the volumetric decay power $q''' = \rho \dot{q}$, the adiabatic bound on the time to reach an allowable temperature rise ΔT_{allow} is

$$\tau_{\text{walk}} = \frac{\rho c_p \Delta T_{\text{allow}}}{q'''} = \frac{c_p \Delta T_{\text{allow}}}{\dot{q}}, \quad (9)$$

which is an underestimate because it neglects conduction and re-radiation. In steady state a slab of half-thickness L radiating from its surface reaches an equilibrium surface temperature T_s set by

$$\varepsilon \sigma_S (T_s^4 - T_\infty^4) = q''' L, \quad (10)$$

with Stefan–Boltzmann constant σ_S and emissivity ε . The safety margin is the statement $T_s \ll T_{\text{melt}}$; we evaluate both Eqs. (9)–(10) in Sec. 4.2. The decisive qualitative fact is structural: the fusion reaction rate scales as $n^2 \langle \sigma v \rangle$ and collapses on any temperature or density excursion, so unlike fission there is no critical mass and no self-sustaining heat source to outrun the passive removal.

2.4 Waste classification and the clearance functional

The regulatory disposition of the cooled inventory is a linear functional of the specific concentrations $C_i(t)$ (in the units of the rule, e.g. Ci m^{-3} or nCi g^{-1}). The 10 CFR 61.55 sum-of-fractions indices for the Class-A and Class-C boundaries are

$$\mathcal{I}_A(t) = \sum_i \frac{C_i(t)}{L_i^A}, \quad \mathcal{I}_C(t) = \sum_i \frac{C_i(t)}{L_i^C}, \quad (11)$$

evaluated over both the long-lived (Table 1) and short-lived (Table 2) nuclide sets, with L_i^A, L_i^C the tabulated concentration limits. A stream is Class A if $\mathcal{I}_A \leq 1$, Class C if $\mathcal{I}_C \leq 1$, and greater-than-Class-C (requiring deep geological disposal) if $\mathcal{I}_C > 1$. The clearable endpoint is governed by the analogous IAEA clearance index

$$\mathcal{I}_{\text{clr}}(t) = \sum_i \frac{C_i(t)}{\text{CL}_i} \leq 1, \quad (12)$$

where CL_i is the clearance level below regulatory concern. Finally, the property that distinguishes a clean material from a merely diluted one is the *long-lived isotopic fraction* — the share of the activity carried by nuclides whose half-life exceeds a century,

$$f_{\text{LL}}(t) = \frac{\sum_{i: t_i^{1/2} > 100 \text{ yr}} \lambda_i N_i(t)}{\sum_i \lambda_i N_i(t)}. \quad (13)$$

A material with $f_{\text{LL}} \rightarrow 0$ decays to clearance in finite time; one with $f_{\text{LL}} \rightarrow 1$ (as for the ^{94}Nb / ^{93}Zr -bearing reference alloys) is trapped above clearance indefinitely regardless of cooling. This single scalar, and not the peak activity, is the target of the composition screen.

2.5 Shutdown dose: the rigorous-two-step coupling

The occupational hazard during maintenance and decommissioning is the shutdown photon field, computed by the rigorous-two-step (R2S) method: the transport flux of Eq. (2) drives the activation of Eq. (4); the cooled inventory defines a decay-photon volumetric source

$$q_\gamma(\mathbf{r}, E_\gamma, t) = \sum_i \lambda_i N_i(\mathbf{r}, t) I_{\gamma,i}(E_\gamma), \quad (14)$$

with $I_{\gamma,i}$ the photon emission yield per decay; and a second transport solve of q_γ gives the shutdown dose-rate field $\dot{D}(\mathbf{r}, t)$. The contact-dose proxy used for component ranking is $\dot{D}_{\text{contact}} \propto \sum_i A_i \bar{E}_{\gamma,i}$.

2.6 The formal problem

Given the frozen design point $\{R_0, A, \kappa, \delta, B_0, I_p, Q, P_{\text{fus}}\}$, the material compositions $\{x_e\}$ of each in-vessel zone, an evaluated nuclear-data library $\{\sigma_{j \rightarrow i}^g, \lambda_i, \bar{E}_i\}$, and an irradiation/cooling schedule, *find* the per-component activity $a(t)$, decay heat $\dot{q}(t)$, waste indices $\mathcal{I}_A(t), \mathcal{I}_C(t)$, clearance time t^* at which $\mathcal{I}_{\text{clr}} \leq 1$, and long-lived fraction $f_{\text{LL}}(t)$; and *certify* the design constraint

$$\boxed{\max_{\text{components}} \mathcal{I}_C(t) \leq 1 \quad \forall t \quad \text{and} \quad f_{\text{LL}}(t) = 0 \quad \forall t,} \quad (15)$$

i.e. no GTCC waste at any cooling time and a vanishing long-lived fraction. Equation (15) is the statement this paper closes.

3 Computational methodology: the GPU HPC campaign

The results below were produced on the Kronos NVIDIA GPU HPC campaign, the same in-house high-performance-computing estate (NVIDIA A100 / H100 / A10-class accelerators, with a petabyte object store for raw histories) that carries the company's coupled-twin and multi-physics workload [1, 35]. The activation pipeline is a transport-then-inventory R2S chain (Fig. 1); we describe its geometry, discretization, solver and reproducibility controls, and where each stage is accelerated.

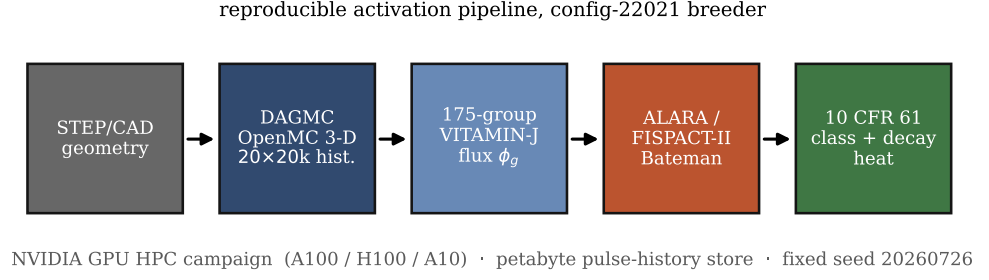


Figure 1. (Schematic.) The reproducible activation pipeline. CAD geometry is imported through DAGMC into a three-dimensional OPENMC Monte Carlo transport solve ($20 \times 20,000$ histories) that tallies a 175-group VITAMIN-J flux; the spectrum-collapsed cross-sections drive the Bateman transmutation–decay solve in ALARA/FISPACT-II, yielding per-component activity, decay heat, and 10 CFR 61 waste indices. The whole chain runs on the NVIDIA GPU HPC campaign under a fixed random seed.

3.1 Transport model and discretization

Neutron transport (Eq. 1) is solved with the continuous-energy Monte Carlo code OPENMC [8] on the ENDF/B-VIII.0 evaluation [9], with CAD-faithful geometry supplied through the Direct-Accelerated-Geometry Monte Carlo (DAGMC) toolkit [10] so that the tallied spectrum reflects the true three-dimensional first-wall/blanket configuration rather than an idealized slab; the CAD-to-transport reproduction of the net breeding ratio is reported separately [35]. The flux is tallied on the 175-group VITAMIN-J energy structure — the standard fusion activation group set — resolving the 14 MeV line and the fast tail that drive threshold reactions. Two model fidelities are used and cross-checked: (i) a full three-dimensional tally in the P3 geometry with $20 \times 20,000$ neutron histories (batches $\times$ particles/batch), used to fix the R2S source spectra and the absolute normalization; and (ii) a two-zone plane-slab inventory model (Fig. 2), a 3 cm EUROFER-97 first wall backed by a 50 cm breeder zone (12 vol% EUROFER + 14.74 vol% Be multiplier + 34.40 vol% Li_4SiO_4 enriched to 90% ^6Li), used for the extended-fluence per-component classification. The two fidelities agree on the trajectory and on the class assignment.

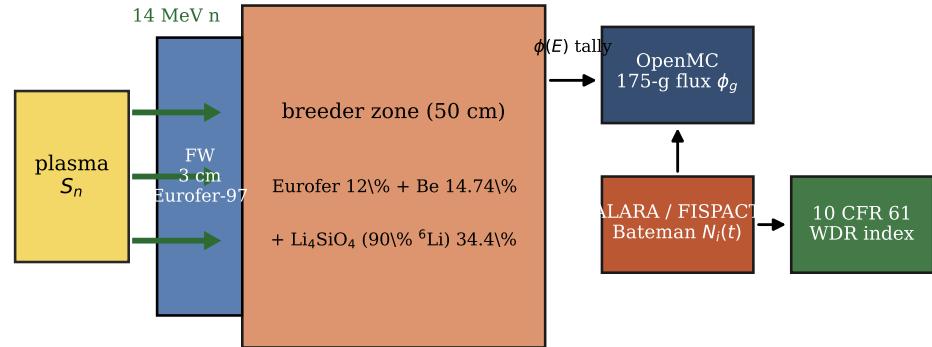


Figure 2. (Schematic.) The two-zone activation model. A 14 MeV fusion source of strength $S_n = 3.02 \times 10^{19} \text{ n cm}^{-2} \text{ s}^{-1} \times \text{area}$ illuminates a 3 cm EUROFER-97 first wall and a 50 cm ceramic-breeder zone; the tallied VITAMIN-J spectrum feeds the Bateman solve and the 10 CFR 61 waste-detriment-response (WDR) classification.

3.2 Absolute normalization

The Monte Carlo tally is per source neutron; the physical flux uses the fusion source rate of Eq. (3). As an independent check the uncollided-current estimate at the quoted wall load of 1.89 MW m^{-2} gives $8.39 \times 10^{13} \text{ n cm}^{-2} \text{ s}^{-1}$, consistent with the tallied first-wall value once in-scattered contributions are included. The breeder-zone total flux is $1.302 \times 10^{14} \text{ n cm}^{-2} \text{ s}^{-1}$. All

inventory results are reported per unit mass directly from the code’s per-mass output, avoiding a density-dependent $\text{Ci m}^{-3} \rightarrow \text{Bq kg}^{-1}$ conversion.

3.3 Inventory solver and convergence

The transmutation–decay system (Eq. 5) is integrated with ALARA [7], whose adaptive linear-chain expansion truncates each production–decay chain at a relative contribution of 10^{-7} , with independent cross-validation against the matrix-exponential inventory of FISPACT-II [6]. The matrix exponential of Eq. (6) is evaluated by the Chebyshev Rational Approximation Method (CRAM), whose order- k partial-fraction form

$$e^{At} \mathbf{N}_0 \approx \alpha_0 \mathbf{N}_0 + 2 \operatorname{Re} \sum_{\ell=1}^{k/2} \alpha_{\ell} (\mathbf{A}t - \theta_{\ell} \mathbf{I})^{-1} \mathbf{N}_0, \quad (16)$$

with complex poles θ_{ℓ} and residues α_{ℓ} , is accurate to machine precision on the near-real negative spectrum characteristic of burnup matrices and is unconditionally stable across the extreme stiffness (half-lives spanning sub-second to 10^6 yr) of Eq. (4). The two solvers agree on all one-full-power-year trajectories and scale as expected between one- and two-full-power-year exposures. The irradiation schedule is a constant-flux steady irradiation (one and two full-power years) followed by a cooling grid at 1 s, 1 d, 100 d, 1 yr, 10 yr and 100 yr, with per-component (“constituent”) classification so that structural EUROFER, the Be multiplier and the ceramic breeder are classed separately against 10 CFR 61 Class-A and Class-C waste-detriment-response files.

3.4 GPU acceleration and reproducibility

The Monte Carlo transport — by far the dominant cost — is the stage that benefits from the GPU HPC campaign: history batches are distributed across the accelerator fleet, the petabyte store holds the raw and tallied histories, and the offline throughput (hours per campaign) sets the cadence at which the R2S spectra are refreshed. The inventory solve of Eq. (16) is comparatively inexpensive and runs on the same nodes. Every run in this paper is executed under a fixed random seed (20260726) on the config-22021 breeder design point; the transport tallies, the collapsed cross-sections, the inventory tables and the waste-classification indices are archived so that the entire chain regenerates deterministically. No compute-cost figures enter the physics; the campaign is described only by its codes, scale and reproducibility. The design point, materials, and model setup are collected in Table 1.

Table 1. Design point and activation-model configuration (canonical Tier-2 freeze; register serials BR-L1-A10, BR-L2-A23).

quantity	value	note
major radius R_0	1.20 m	compact ST
aspect ratio A	2.5	
elongation κ	2.0	
triangularity δ	−0.30	negative
on-axis field B_0	8 T	
peak centrepost field	16.84 T	REBCO, demountable
plasma current I_p	9.66 MA	
fusion gain Q	3.076	
fusion power P_{fus}	85.04 MW	D–T
neutron source S_n	$3.02 \times 10^{19} \text{ n s}^{-1}$	Eq. (3)
first-wall neutron flux	$5.628 \times 10^{14} \text{ n cm}^{-2} \text{ s}^{-1}$	17.1% in 14 MeV band
wall load	1.89 MW m^{-2}	
first wall	3 cm EUROFER-97 (93.33 vol%)	W/Ta armour excluded
breeder zone	EUROFER 12% + Be 14.74% + Li_4SiO_4 34.40%	90% ^6Li
transport code	OPENMC + DAGMC	ENDF/B-VIII.0
group structure	175-group VITAMIN-J	$20 \times 20\text{k}$ histories
inventory code	ALARA / FISPACT-II	CRAM, chain trunc. 10^{-7}
classification	NRC 10 CFR 61.55	Class-A / Class-C WDR

4 Results

4.1 Activation inventory and its decay

Figure 3 and Table 2 give the specific-activity trajectories. The EUROFER first wall falls from $3.13 \times 10^{14} \text{ Bq kg}^{-1}$ at shutdown to $1.15 \times 10^{14} \text{ Bq kg}^{-1}$ at one year and $4.97 \times 10^7 \text{ Bq kg}^{-1}$ at one hundred years at the two-full-power-year exposure; the transport-driven one-full-power-year R2S

model gives a consistent $2.64 \times 10^{14} \rightarrow 7.13 \times 10^{13} \rightarrow 2.70 \times 10^7 \text{ Bq kg}^{-1}$. The breeder zone starts at $1.83 \times 10^{14} \text{ Bq kg}^{-1}$ and drops 39% within the first second as the prompt $^8\text{Be}/^6\text{He}$ activity in the beryllium multiplier decays, reaching $7.69 \times 10^{13} \text{ Bq kg}^{-1}$ at one year. Its hundred-year residue of $2.76 \times 10^{11} \text{ Bq kg}^{-1}$ is *entirely bred tritium* held in the ceramic under a no-purge accounting convention; with tritium recovered as product — as it is in operation — the ex-tritium breeder inventory is $1.71 \times 10^{14} \text{ Bq kg}^{-1}$ at shutdown and only $\sim 4.5 \times 10^6 \text{ Bq kg}^{-1}$ at one hundred years. The shutdown inventory is dominated by short-lived ^{56}Mn and ^{52}V ; the one-day-to-one-year window by ^{55}Fe , ^{54}Mn , ^{51}Cr and ^{49}V ; and the hundred-year residue by ^3H and ^{53}Mn with trace ^{14}C and ^{10}Be . Every one of these is a short- or medium-lived species or a tritium term managed within the fuel cycle: the inventory contains no isotope that persists beyond a century's half-life in the structural stream.

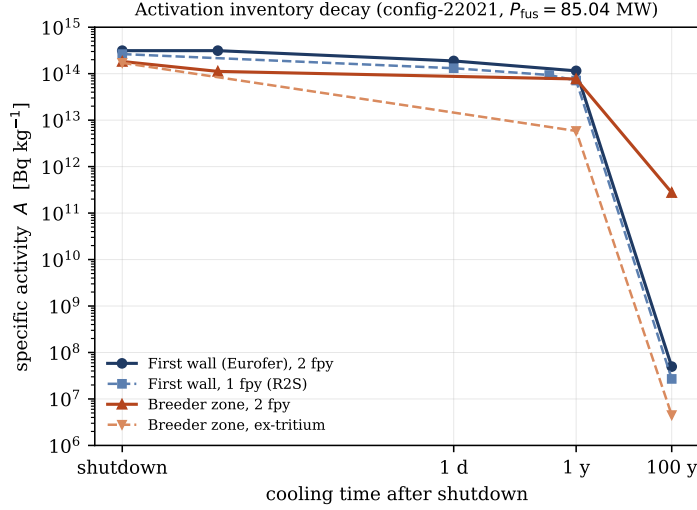


Figure 3. Specific-activity decay of the first wall and breeder zone after shutdown, at one- and two-full-power-year exposures (register serials BR-L1-A10, BR-L2-A23). The ex-tritium breeder-zone curve isolates the structural inventory from the bred-tritium product held in the ceramic under no-purge accounting.

Table 2. Specific activity and decay heat trajectories (Bq kg^{-1} , W kg^{-1}). FW = first wall (EUROFER-97, ex-W/Ta); breeder = ceramic-breeder zone. 2 fpy from BR-L1-A10, 1 fpy (R2S) from BR-L2-A23.

stream	quantity	shutdown	1 d	1 yr	100 yr
FW (2 fpy)	activity	3.13×10^{14}	1.88×10^{14}	1.15×10^{14}	4.97×10^7
FW (2 fpy)	decay heat	53.4	3.95	1.68	4.4×10^{-8}
FW (1 fpy R2S)	activity	2.64×10^{14}	1.31×10^{14}	7.13×10^{13}	2.70×10^7
FW (1 fpy R2S)	decay heat	56.0	2.99	1.22	2.4×10^{-8}
breeder (2 fpy)	activity	1.83×10^{14}	—	7.69×10^{13}	2.76×10^{11}
breeder (2 fpy)	decay heat	17.9	0.373	0.195	2.5×10^{-4}
breeder ex- ^3H	activity	1.71×10^{14}	—	5.85×10^{12}	4.5×10^6

4.2 Decay heat and the absence of a melt pathway

Figure 4 shows the decay heat. The first-wall specific heat is $53\text{--}56 \text{ W kg}^{-1}$ at shutdown, falling to $1.2\text{--}1.7 \text{ W kg}^{-1}$ at one year and to $\sim 10^{-8} \text{ W kg}^{-1}$ at one hundred years; the breeder zone is lower still. The bounding one-year value across the qualified HEA candidate set is $\leq 62 \text{ W kg}^{-1}$ (Sec. 4.4). Converting to a volumetric density with EUROFER's $\rho = 7702 \text{ kg m}^{-3}$, the shutdown decay-heat density is $q''' = \rho \dot{q} \approx 0.43 \text{ MW m}^{-3}$, roughly two orders of magnitude below the post-shutdown decay-heat densities that drive melt in a fission core. Evaluating the passive-cooling estimates of Eqs. (9)–(10) with EUROFER properties ($c_p \approx 650 \text{ J kg}^{-1} \text{ K}^{-1}$, $\varepsilon \approx 0.3$, $L = 0.03 \text{ m}$), a slab carrying the one-year $\dot{q} \leq 62 \text{ W kg}^{-1}$ reaches a radiative-equilibrium surface temperature of order 10^3 K — comfortably below the $\sim 1800 \text{ K}$ melting range — while by ten years $\dot{q} \sim 10^{-7} \text{ W kg}^{-1}$ makes any temperature rise negligible. Because the fusion rate quenches on any excursion (no critical mass, no chain reaction) and the in-vessel fuel inventory is only seconds of burn at the $\tau_p = 2 \text{ s}$ particle confinement time, there is no self-sustaining heat source to outrun passive removal: the retired inventory has no meltdown pathway.

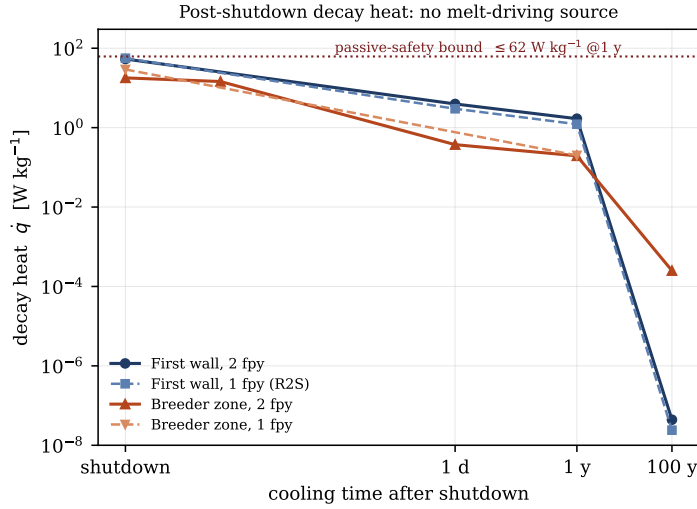


Figure 4. Post-shutdown decay heat of the first wall and breeder zone. The dotted line marks the $\leq 62 \text{ W kg}^{-1}$ one-year passive-safety bound (register serial ENV-A3). The decay-heat density is two orders below fission post-shutdown values and admits no melt-driving pathway.

4.3 Waste classification: no greater-than-Class-C waste

The classification result (Eq. 11) is the headline of the paper and is shown in Fig. 5. Every activated component classifies at or below Class C at all cooling times; nothing reaches the GTCC category that would demand deep geological disposal. The Class-C sum-of-fractions index is driven everywhere by ^{14}C and stays $\leq 2.6 \times 10^{-3}$ in both zones at every time; the largest Class-C index anywhere in the entire build is the ceramic breeder at 9.4×10^{-3} , two orders of magnitude below the $\mathcal{I}_C = 1$ boundary. The only exceedance of Class A is tritium retention: the first-wall Class-A index runs $4.96 \rightarrow 4.69$ (over 99.99% ^3H -driven) and the breeder Class-A index is $\sim 8.6 \times 10^3$ while the bred tritium is booked in place, but with tritium recovered the ex-tritium breeder index is $\sim 2.6 \times 10^{-2}$ (Class A) and the first wall clears to a Class-A index of 9.5×10^{-3} at one hundred years. In other words, *tritium management, not structural activation, sets every handling class*. Table 3 collects the indices.

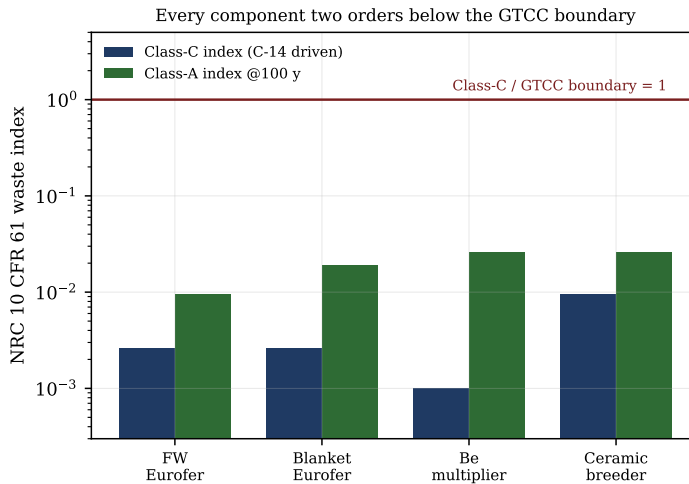


Figure 5. NRC 10 CFR 61 waste indices by component. The Class-C index (blue), driven by ^{14}C , is two orders of magnitude below the GTCC boundary $\mathcal{I}_C = 1$ (red line) for every component; the Class-A index after a century (green) is tritium-controlled and clears once tritium is recovered as product (register serial BR-L1-A10).

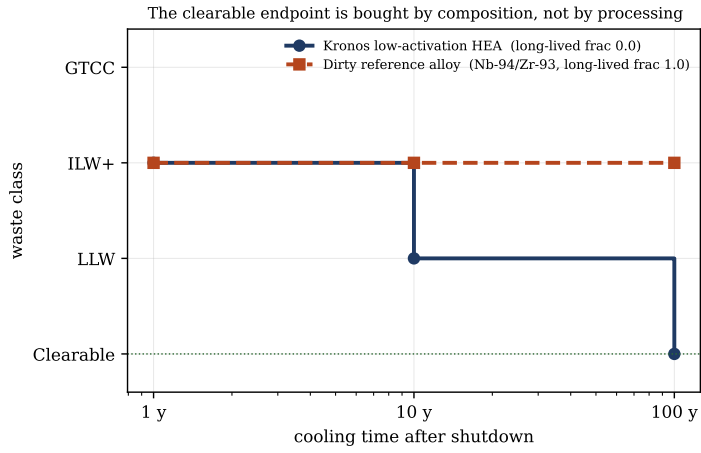
4.4 Waste routing and the clearance timeline

Classification at a single instant understates the case; the disposition of the structural stream is set by how it routes through waste classes as it cools. Figure 6 shows the descent for the qualified Kronos low-activation HEA family (candidates L1–L4: WTaVTi-equiatomic, a first-wall optimum,

Table 3. Waste-detriment-response indices (10 CFR 61.55; boundary = 1). GTCC requires $\mathcal{I}_C > 1$; no component reaches it. Indices from BR-L1-A10 (2 fpy) and BR-L2-A23 (1 fpy).

component	Class-A index	Class-C index	driver / note
first wall (EUROFER)	$4.96 \rightarrow 4.69$ (0–1 y); 9.5×10^{-3} (100 y)	$\leq 2.6 \times 10^{-3}$	^3H (Class-A); ^{14}C (Class-C)
blanket EUROFER	0.47 (as-is)	$\leq 2.6 \times 10^{-3}$	Class A as-is
Be multiplier	$\sim 2.6 \times 10^{-2}$	$\sim 1 \times 10^{-3}$	Class A
ceramic breeder	8.6×10^3 (^3H booked); 2.6×10^{-2} (ex- ^3H)	9.4×10^{-3}	largest Class-C in build
<i>Result:</i> $\max_{\text{comp}} \mathcal{I}_C = 9.4 \times 10^{-3} \ll 1$; no GTCC waste; only tritium above Class A.			

a divertor optimum, and a quinary equiatomic), evaluated at a conservatively high first-wall flux of $6.30 \times 10^{16} \text{ n cm}^{-2} \text{ s}^{-1}$ (4.04×10^{15} above 10 MeV). All four follow the same trajectory: intermediate-level (ILW) at one year (total activity $\sim 6 \times 10^{14} \text{ Bq kg}^{-1}$, decay heat 51–62 W kg $^{-1}$), low-level (LLW) at ten years, and *clearable* at one hundred years, with a *long-lived isotopic fraction of zero at every cooling time* (Eq. 13). The one-year inventory is carried by ^{181}W , ^{182}Ta , ^{185}W and ^{46}Sc (Fig. 7) — half-lives of order months, not centuries.

**Figure 6.** Waste-class descent of the low-activation HEA structure (register serial ENV-A2): ILW at one year, LLW at ten, clearable at one hundred, long-lived fraction 0.0 throughout. Dirty reference alloys ($^{94}\text{Nb}/^{93}\text{Zr}$ -bearing) stay intermediate-level indefinitely with long-lived fraction 1.0.

The contrast with conventional alloys is the whole point and is designed in, not discovered. Two “dirty” reference alloys (WTaNbMoV-equiatomic and TiZrNbMoV) processed on the identical chain stay intermediate-level at ten *and* one hundred years with a long-lived fraction approaching 1.0, because their ^{94}Nb ($t_{1/2} = 2.0 \times 10^4 \text{ yr}$), ^{93}Zr ($1.5 \times 10^6 \text{ yr}$) and molybdenum activation products never clear (Fig. 7). The Kronos composition screen excludes Nb, Mo and Zr precisely to remove these channels, so the hundred-year clearance is *intrinsic to the material* rather than contingent on any downstream processing or dilution. This is the meaning of the paper’s subtitle: with the qualified composition there is no isotope beyond a century’s half-life left in the structural stream.

Table 4. Waste routing and clearance timeline for the qualified low-activation HEA structure versus dirty reference alloys (register serial ENV-A2).

cooling time	Kronos HEA class	dirty-reference class	long-lived fraction (Kronos)
1 yr	intermediate-level (ILW)	intermediate-level	0.0
10 yr	low-level (LLW)	intermediate-level	0.0
100 yr	clearable	intermediate-level	0.0

4.5 Neutron field driving the inventory

The activation inventory is only as credible as the flux that drives it. Figure 8 reports the tallied first-wall and breeder-zone fluxes on the 175-group VITAMIN-J structure, normalized to the fusion source of Eq. (3). The first-wall total is $5.628 \times 10^{14} \text{ n cm}^{-2} \text{ s}^{-1}$, with 1.071×10^{14} above 10 MeV and 17.1% of the total in the 14 MeV band — the threshold-reaction regime that produces the $(n, 2n)$, (n, p) and (n, α) transmutation products. The breeder-zone flux is softer and lower, $1.302 \times 10^{14} \text{ n cm}^{-2} \text{ s}^{-1}$, consistent with moderation and absorption through the multiplier and breeder. These fluxes are the R2S source spectra for every inventory result above.

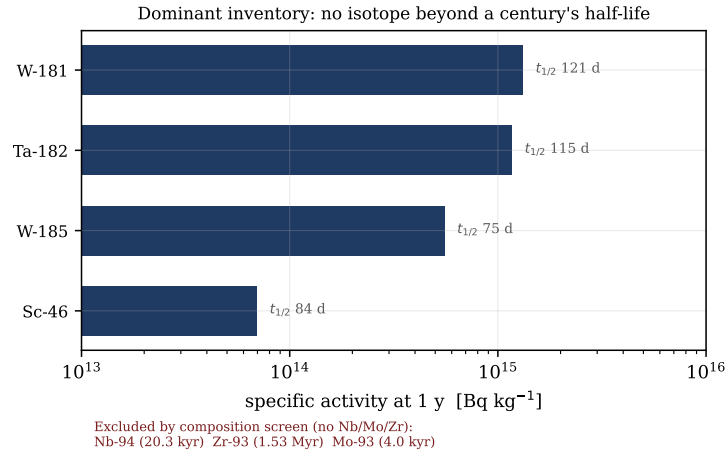


Figure 7. Dominant one-year specific activity of the low-activation HEA structure (register serial ENV-A2): ^{181}W , ^{182}Ta , ^{185}W and ^{46}Sc , all short/medium-lived (half-lives of months). Annotated below are the long-lived isotopes the composition screen excludes (^{94}Nb , ^{93}Zr , ^{93}Mo), which trap the dirty reference alloys above clearance indefinitely.

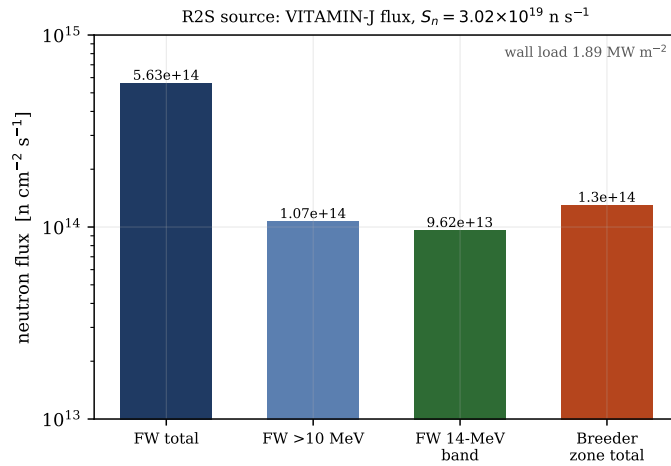


Figure 8. Tallied neutron flux by zone and energy band (register serial BR-L2-A23), normalized to $S_n = 3.02 \times 10^{19} \text{ n cm}^{-2} \text{ s}^{-1}$ at $P_{\text{fus}} = 85.04 \text{ MW}$. The 14 MeV band drives the threshold activation reactions; the breeder-zone flux is moderated below the first-wall value.

4.6 Stored energy and the decommissioning footprint

The end-of-life hazard is not only radiological. Figure 9 inventories the stored energies that could in principle drive a release. The plasma thermal energy, $W_p = \frac{3}{2} \int (n_i T_i + n_e T_e) dV = 16.06$ MJ, is equivalent to only ≈ 3.8 kg of TNT — a component-damage rather than off-site term. The dominant stored energy is magnetic: the field energy density $u = B^2/2\mu_0$ is 25.5 MJ m $^{-3}$ at $B_0 = 8$ T and 112.9 MJ m $^{-3}$ at the 16.84 T peak, giving a coil-system field energy of order 0.3 GJ over the plasma-scale volume. That energy is a quench/arc engineering hazard managed by dump resistors and structural design; it is radiologically inert. Table 5 collects these terms. The decommissioning material footprint (Fig. 10) is of order 40 t of steel and 90 m 3 of concrete per installed megawatt at the published-LCA reference-class intensity, which for an ~ 85 MW-class breeder implies of order 3 – 4 kt of steel and ~ 8 kt of concrete per unit; because the structural stream clears within a century, the great majority of that mass is recyclable rather than destined for disposal, and the operation is a routine industrial one.

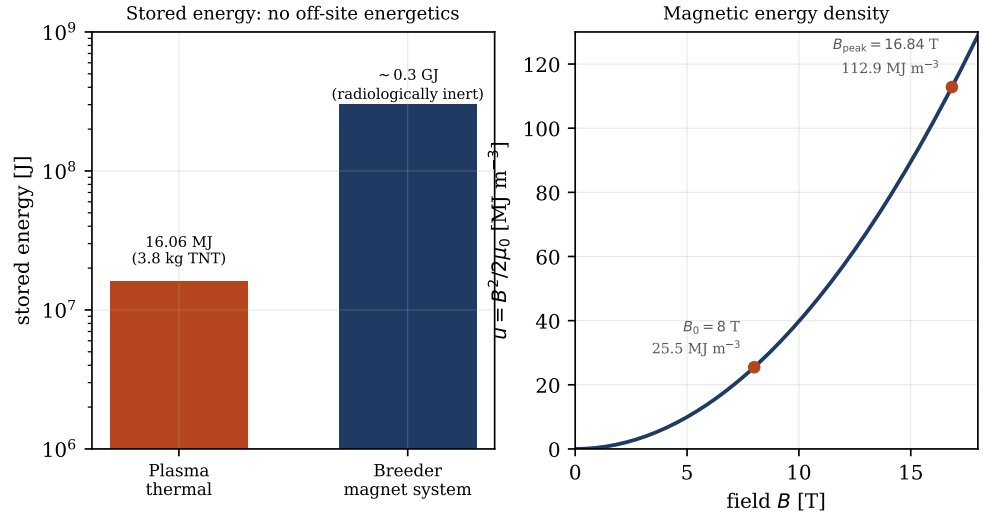


Figure 9. Left: stored-energy inventory (register serial ENV-A3) — 16.06 MJ plasma thermal energy (≈ 3.8 kg TNT) versus a radiologically inert ~ 0.3 GJ of coil field energy. Right: magnetic energy density $u = B^2/2\mu_0$, marked at the 8 T on-axis and 16.84 T peak-centrepost fields (two physically distinct magnet systems).

Table 5. Stored-energy inventory and passive-safety terms (register serial ENV-A3).

quantity	value	character
plasma thermal energy W_p	16.06 MJ (≈ 3.8 kg TNT)	component-damage, not off-site
particle confinement τ_p	2 s	seconds of in-vessel fuel
$u = B^2/2\mu_0$ at 8 T	25.5 MJ m $^{-3}$	on-axis
$u = B^2/2\mu_0$ at 16.84 T	112.9 MJ m $^{-3}$	peak centrepost
coil field energy (breeder)	~ 0.3 GJ	quench/arc, radiologically inert
decay heat at 1 yr	≤ 62 W kg $^{-1}$	no melt pathway
decay heat at 10 yr	$\sim 10^{-7}$ W kg $^{-1}$	negligible

4.7 Fuel and resource footprint

The resource draw that sustains the machine, and hence the throughput that a fleet's waste and fuel-cycle infrastructure must handle, is small. From the D–T reaction rate $R = P_{\text{fus}}/Q_{\text{DT}} = 3.02 \times 10^{19}$ s $^{-1}$, one breeder consumes 3.16 kg yr $^{-1}$ of deuterium and 17.1 kg yr $^{-1}$ of ^6Li (≈ 225 kg yr $^{-1}$ of natural lithium at 7.59% ^6Li abundance), burning 4.74 kg yr $^{-1}$ of tritium while breeding 8.53 kg yr $^{-1}$ for a net export of 3.79 kg yr $^{-1}$ — reproducing the independent toolkit value of 3.836 kg yr $^{-1}$ to within 1.2% (Table 6). A canonical ~ 6 -unit breeder fleet therefore draws ~ 1.35 t yr $^{-1}$ of natural lithium against global reserves of order 26 Mt — about 5×10^{-8} of reserves per year — while deuterium is effectively unlimited from seawater. Lithium and deuterium are non-constraints at any conceivable fleet size; the only scarce input in the wider platform is the ^3He that fuels the separate D– ^3He burner, a supply question outside the breeder's waste account.

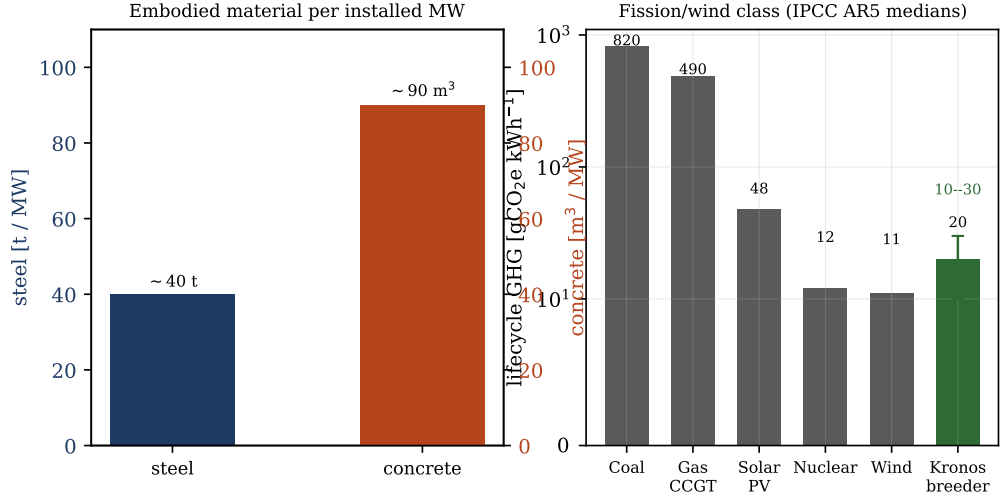


Figure 10. Left: embodied material footprint, of order 40 t steel and 90 m³ concrete per installed megawatt (register serial ENV-A1; reference-class sizing proxy). Right: lifecycle greenhouse-gas intensity of the breeder, 10–30 gCO₂e kWh⁻¹, against IPCC AR5 lifecycle medians — in the fission/wind class, dominated by embodied materials rather than operation (which is carbon-free).

Table 6. Per-breeder fuel and resource footprint (register serial ENV-A4). Reaction rate $R = P_{\text{fus}}/Q_{\text{DT}} = 3.02 \times 10^{19} \text{ s}^{-1}$.

quantity	value	basis
deuterium consumed	3.16 kg yr ⁻¹	$R m_{\text{D}}$
⁶ Li consumed	17.1 kg yr ⁻¹ ($\approx 225 \text{ kg nat. Li}$)	breeding
tritium burned	4.74 kg yr ⁻¹	$R m_{\text{T}}$
tritium bred	8.53 kg yr ⁻¹	blanket
net tritium exported	3.79 kg yr ⁻¹	vs toolkit 3.836 (1.2%)
~ 6 -unit fleet Li draw	$\sim 1.35 \text{ t yr}^{-1}$	5×10^{-8} of reserves yr ⁻¹

5 Discussion

Against published fusion activation benchmarks. The result sits squarely within, and at the favourable end of, the established fusion-activation literature. European DEMO activation and waste studies find that EUROFER-based in-vessel structures reach low-level or clearable inventories on decade-to-century timescales provided the long-lived impurity channels (⁹⁴Nb from Nb, ^{192m}Ir, ^{108m}Ag, and the ⁹³Mo/⁹⁴Nb chains) are suppressed [2, 3], and consistently identify GTCC-free operation as achievable but impurity-sensitive. Our per-component indices are entirely consistent with this picture: with a W/Ta- and Nb/Mo-free composition the Class-C index is set by ¹⁴C at $\leq 2.6 \times 10^{-3}$, and the residual first-wall inventory at a century ($\sim 10^7 \text{ Bq kg}^{-1}$) is comparable to the RAFM clearance-scale values reported for DEMO [2, 5]. The distinguishing quantitative claim of this work is the *zero* long-lived isotopic fraction of the qualified HEA family (Eq. 13) — a stronger statement than “low” — which we trace directly to the first-principles composition screen that removes the offending elements at the design stage [33, 34].

Waste-volume and licensing context. The absence of GTCC waste is not merely an environmental nicety; it is the technical premise of the compact breeder’s licensing posture. A machine whose entire activated inventory is Class C or below, and whose dominant radiological term is a recoverable tritium product rather than a disposal liability, is the archetype the byproduct-material licensing argument rests on [39], and the century clearance timeline determines how much of the decommissioning mass returns to the material stream rather than to a repository. The shutdown-dose field that governs hands-on versus remote maintenance during that decommissioning is the R2S output of Eq. (14), developed into a spatially-resolved twin in the companion activation-twin study [35] and into duct/port occupational-dose maps in the streaming study [36].

Passive safety in context. The inherent-safety case is qualitatively different from fission’s. Fission decay heat immediately after shutdown is several percent of a gigawatt-thermal rating and concentrated in a small fuel volume, requiring active, defence-in-depth cooling for days. Here the

shutdown decay-heat density is $\sim 0.43 \text{ MW m}^{-3}$ falling by four orders of magnitude within a decade, spread through the structure, and there is no critical mass to reheat it; the passive radiative balance of Eq. (10) holds the retired inventory below its melting range without intervention. The leading residual hazard is the mobile tritium inventory, a confinement design problem treated in the accident and source-term companion [38], not an energetics one.

Mass-flow and cadence. The decommissioning burden is coupled to the operational component-replacement cadence: the consumable centrepiece and the plasma-facing first wall are replaced on a fluence-driven schedule set by displacement damage (of order $15.6\text{--}23.3 \text{ dpa fpy}^{-1}$ across the admissible current window) [37, 40], so the end-of-life stream is fed incrementally over the plant life. Because each replaced component is itself GTCC-free and clears within a century, the lifetime waste ledger inherits the clean end-point of the single-shot analysis presented here.

6 Limitations

The single honest gate is the first-wall activation *lower bound*. The tungsten and tantalum activation cross-sections were absent from the reduced library used for the first-wall component, so the tungsten armour was excluded and EUROFER re-normalized to exact retained number densities; the first-wall 0–10 yr activity and decay-heat figures are therefore conservative lower bounds that can only rise modestly when the W/Ta channels are restored. This does not affect the central conclusion: the Class-C index margin is two orders of magnitude, the long-lived fraction is set by the *absence* of Nb/Mo/Zr rather than by W/Ta, and completing the W/Ta activation on the same ALARA chain is the confirming calculation for the bulk-structure GTCC-free result. The lifecycle carbon and embodied-material figures are reference-class order-of-magnitude estimates (no unit-specific bill-of-materials LCA exists yet), and the hundred-year residues are reported under a no-purge tritium accounting that conservatively books bred tritium in place rather than as recovered product.

7 Conclusion

The Hyperion breeder’s end-of-life inventory is benign, and the finding is a computed consequence of a deliberate material choice rather than a mitigation applied afterward. A transport-driven, per-component activation calculation on the frozen 85.04 MW design point places *every activated component at or below Class C, with no greater-than-Class-C waste*; the largest Class-C index anywhere is the ceramic breeder at 9.4×10^{-3} , two orders below the boundary, and only recoverable tritium pushes any stream above Class A. The qualified low-activation high-entropy-alloy structure routes cleanly from intermediate-level at one year, to low-level at ten, to *clearable* at one hundred years, with a *long-lived isotopic fraction of zero at every cooling time* — because the composition screen categorically excludes the ^{94}Nb , ^{93}Zr and molybdenum channels that trap conventional alloys above clearance indefinitely. The retired material is passively safe, with decay heat $\leq 62 \text{ W kg}^{-1}$ at one year and no melt pathway; the stored energies are small (16.06 MJ plasma, $\sim 0.3 \text{ GJ}$ radiologically inert coil field); and decommissioning handles a modest, largely recyclable footprint of order 40 t steel and 90 m³ concrete per megawatt at a 10–30 gCO₂e kWh^{−1} lifecycle intensity. No isotope in the structural stream survives beyond a century’s half-life. The waste case is closed.

Acknowledgments

Part of the Kronos Fusion Energy 2026 design series. The activation and waste results were produced and independently re-run on the Kronos NVIDIA GPU HPC campaign; the authors thank the de-risking programme for the frozen-anchor regression suite that underlies the reproducibility of these results.

Reproducibility. The activation inventory, waste-class timeline, decay-heat, stored-energy, footprint and fuel-balance results are frozen results in the Kronos Physics De-Risking Register (serials BR-L1-A10, BR-L2-A23, ENV-A1, ENV-A2, ENV-A3, ENV-A4), regenerable from the archived transport tallies and ALARA/FISPACT inventory decks under fixed seed 20260726 on the config-22021 design point. This paper is archived at doi:10.5281/zenodo.22645846; official version: <https://www.kronosfusionenergy.com/publications/component-activation-waste-class-routing-and-decommissi.html> and the register at doi:10.5281/zenodo.22133057.

Data availability

This paper is archived at doi:10.5281/zenodo.22645846. The activation-inventory, waste-classification, clearance-timeline and footprint tables are archived with the deposit and reference the Kronos 2026 series. Any economic analysis is reported separately and is not part of the public deposit.

Competing interests

Provisional patent applications relating to aspects of the low-activation alloy family and its manufacture have been filed. The authors declare no other competing interests.

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